



50 QUESTIONS

JEE ADVANCED

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50 QUESTIONS (JEE Advanced Physics)

This file is prepared by Mohit Goenka, Founder of Eduniti and Alumnus of IIT Kharagpur. It is prepared with an aim to help JEE Advanced Aspirants to not just solve good selected questions but to also learn through its video solutions created by Mohit Goenka himself.

Many a times aspirants finds solving JEE Advanced questions a little difficult. These 50 Questions and there solution will help them to build analytical thinking.

These questions have been taken from various sources like AITS, Pathfinder, Foreign books etc.

Categories of questions:

1. Single Correct Questions – S
2. More than one correct Questions – M
3. Numerical Type Questions – N
4. Paragraph Type Questions - P



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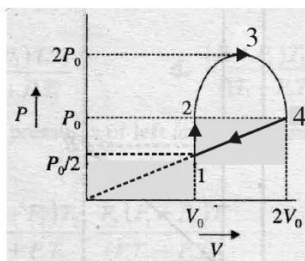
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THERMODYNAMICS

1. Two moles of an ideal monatomic gas undergoes a cyclic process as shown in figure. The path of the process $2 \rightarrow 3 \rightarrow 4$ is elliptical.

1. Heat given to the system in elliptical process is

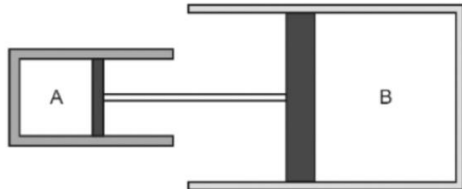
- (a) $\frac{P_0 V_0}{2} \left(7 - \frac{\pi}{4} \right)$ (b) $\frac{P_0 V_0}{3} \left(7 + \frac{\pi}{4} \right)$
(c) $\frac{P_0 V_0}{2} \left(7 + \frac{\pi}{4} \right)$ (d) $\frac{P_0 V_0}{3} \left(7 - \frac{\pi}{4} \right)$



2. Total heat rejected in one cycle is

- (a) $\frac{P_0 V_0 (32 - \pi)}{4}$ (b) $\frac{P_0 V_0 (32 + \pi)}{4}$
(c) $\frac{P_0 V_0 (32 + \pi)}{8}$ (d) $\frac{P_0 V_0 (32 - \pi)}{8}$

2. Two fixed and horizontal cylinders A and B having pistons (both massless) of cross sectional area 100 cm^2 and 200 cm^2 respectively, are connected by massless rod. The piston can move freely without friction. The cylinder A contains 100 gms of an ideal gas ($\gamma = 1.5$) at pressure 10^5 N/m^2 and temperature T_0 . The cylinder B contains identical gas at same temperature T_0 but has different mass. The piston are held at the state such that volume of gas in cylinder A and cylinder B are same and is equal to 10^{-2} m^3 . The walls and piston of cylinder A are thermally insulated where as gas in cylinder B is maintained at constant temperature T_0 . The whole system is in vacuum. Now the pistons are slowly released and they move towards left and mechanical equilibrium is reached at the state when the volume of gas in cylinder A becomes $25 \times 10^{-4} \text{ m}^3$.



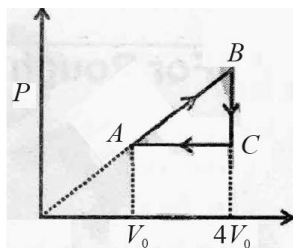
The mass of gas in cylinder B is

- (A) 200 gms (B) 600 gms (C) 500 gms (D) 1 kg

The change in internal energy of gas in cylinder A is :

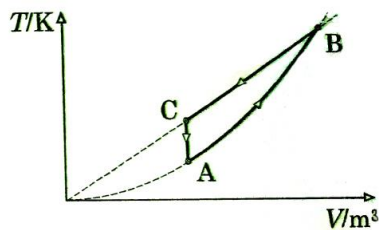
- (A) 2000 J (B) 1000 J (C) 500 J (D) 3000 J

3. A monatomic gas follows a cyclic process $ABCA$ as shown in fig. If the heat given to the gas during process AB is 20J, the heat extracted from the gas during process CA (in Joules) is



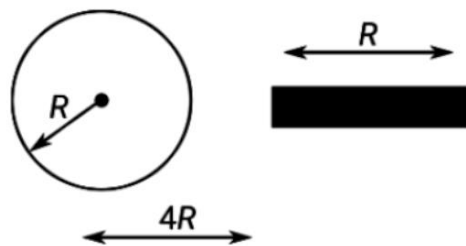
- (a) 6 (b) 10
(c) 5 (d) 8

4. One mole of a mono-atomic ideal gas is made to undergo a quasi-static cyclic process. During the process $A \rightarrow B$, temperature T of the gas varies with its volume V according to the equation $T = bV^2$, where b is a constant. During the whole cycle, maximum pressure of the gas is two times of the minimum pressure. If the gas absorbs 120 J of heat during process $A \rightarrow B$, how much heat does the gas liberate during the process $B \rightarrow C \rightarrow A$?



GRAVITATION

5. A heavy solid sphere with a mass of M and a radius of R is given. A uniform rod with a length of R and a mass of m ($m \ll M$) is kept at rest so that its nearest end is at a distance of $4R$ from the center of the sphere, as shown in the figure below. When the rod is released, it strikes the sphere with a velocity of

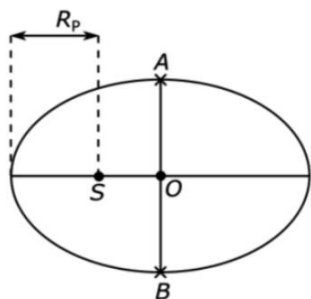


- ☐ A $\left[\frac{2GM}{R} \ln 1.6 \right]^{\frac{1}{2}}$
☐ B $\left[\frac{GM}{2R} \ln 1.6 \right]^{\frac{1}{2}}$
- ☐ C $\left[\frac{2GM}{R} \ln 3.625 \right]^{\frac{1}{2}}$
☐ D $\left[\frac{GM}{2R} \ln 3.625 \right]^{\frac{1}{2}}$

6. A comet orbits the sun with a period of 2.4×10^9 s. Its distance of closest approach, R_p (perihelion distance) to the sun equals 9×10^{10} m. If the time taken by the comet to go round from one end A of the minor axis of an elliptical orbit to the other end B is T_1 and T_2 for clockwise and anticlockwise revolutions respectively, then,

$$\frac{GM}{4\pi^2} = 0.34 \times 10^{19} \text{ SI units,}$$

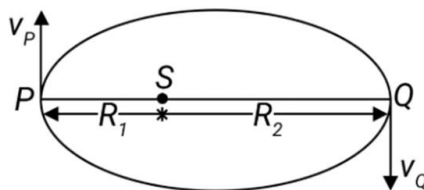
M -Mass of sun, G -Gravitational constant, $(20)^{\frac{1}{3}} = 2.7$ and $\frac{4mb \times 10^{12}}{L} = \text{K m}^{-1} \text{ s}$, m -Mass of the comet; b -semi minor axis, L -Angular momentum about sun]



- A $T_1 - T_2 = 2.79 \text{ Ks}$ B $T_2 - T_1 = 2.61 \text{ Ks}$
C $T_1 - T_2 = 2.61 \text{ Ks}$ D $T_2 - T_1 = 2.79 \text{ Ks}$

7. A comet moving in an elliptical orbit around the Sun S at one of its foci has a speed v_p at a point P on the orbit, as shown in the figure. If $v_p = 1.2v_c$, where v_c is the speed of the comet in circular orbit of radius R_1 around the sun, then ratio of R_1 and R_2 is

- (A) 4:13 (B) 7:18
(C) 5:16 (D) 2:7



8. A star has uniform spherical distribution of matter. The star's mass is m and volume is V . If P_g is the gravitational pressure at centre then

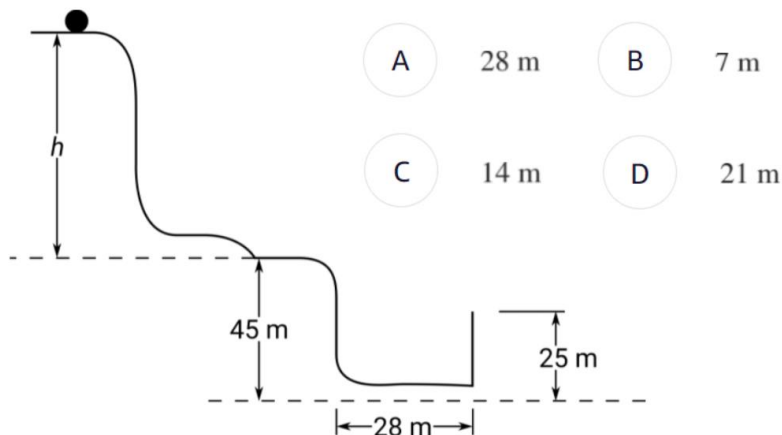
- (A) $P_g \propto V^{-4/3}$ (B) $P_g \propto V^{-1}$
(C) $P_g \propto V^{-2/3}$ (D) $P_g \propto V$

9. A satellite of mass $2m$ is in a circular orbit of radius $2R$ around earth. By mistake, another satellite of mass m is put in the same orbit and having opposite sense of rotation. The collision between the two satellites is perfectly inelastic. Which of the following are correct? (M is mass of earth and R its radius)

- (a) After collision path followed is elliptical.
(b) The combined mass hit the earth surface at an angle $\theta = \cos^{-1} 2/\sqrt{19}$ with the horizontal.
(c) The combined mass does not hit the earth surface and minimum distance from center of earth is $5R/4$.
(d) If radius of earth were $2R/17$ with its mass being same, combined mass would just graze the earth surface.

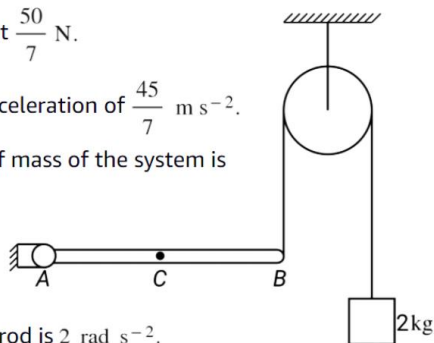
ROTATIONAL MOTION

10. A uniform sphere rolls down the path without slipping, as shown in the figure below. On its path, it encounters a deep pit with a width of 28 m. Now, the minimum height h (to nearest integer) required for the sphere not to fall into the pit should be about (Consider that $g = 10 \text{ m/s}^2$.)



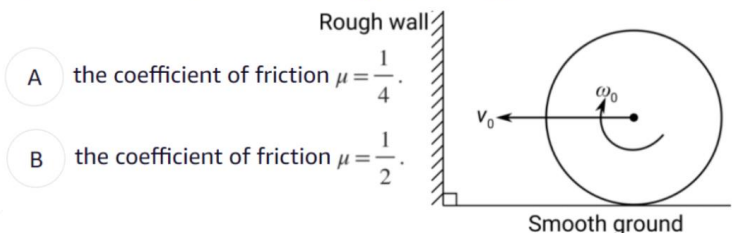
11. A uniform rod AB with a mass of 1 kg and a length of 1 m is pivoted at end A so that the rod can be rotated in vertical plane about an axis through A . The rod is suspended from end B by an inextensible string passing through a fixed and smooth light pulley, as shown in the figure below. The string carries a block with a mass of 2 kg. Initially, the rod is held horizontally and the system is released from rest. Then immediately after release, (Consider that $g = 10 \text{ m/s}^2$.)

- A the tension in the string is about $\frac{50}{7} \text{ N}$.
- B the block goes down with an acceleration of $\frac{45}{7} \text{ m/s}^2$.
- C the acceleration of the center of mass of the system is $\frac{45}{14} \text{ m/s}^2$.



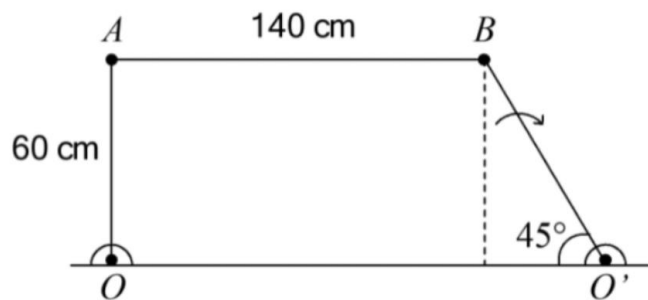
- D the angular acceleration of the rod is 2 rad/s^2 .

12. A solid sphere of mass 2 kg and radius 1 m has a linear velocity $v_0 = 4 \text{ m/s}$ and an angular velocity $\omega_0 = 9 \text{ rad/s}$, as shown. It collides elastically with a rough wall, with a coefficient of friction μ . After colliding with the wall, the sphere immediately rolls without slipping in the opposite direction. Then



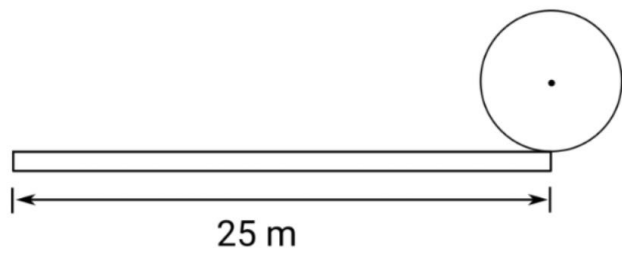
- A the coefficient of friction $\mu = \frac{1}{4}$.
- B the coefficient of friction $\mu = \frac{1}{2}$.
- C the impulse imparted to the sphere by the ground surface during impact is 2 N s .
- D the impulse imparted to the sphere by the ground surface during impact is 4 N s .

13. The arm $O'B$ of the linkage has a clockwise angular velocity of 10 rad s^{-1} in the position shown in the figure. What is the velocity of A in ms^{-1} ?

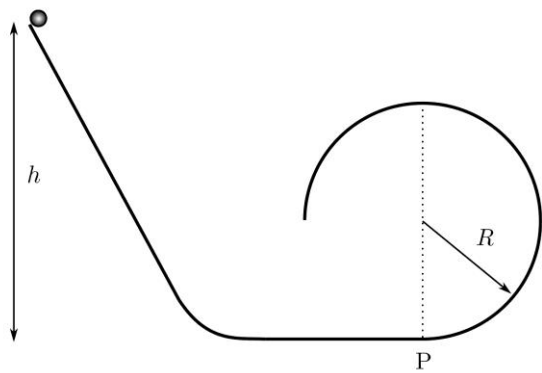


(here linkage OA and $O'B$ are pivoted at O and O' and can rotate about O and O')

14. A solid cylinder is placed on one edge of a plank of same mass and length 25 m placed on a smooth surface, as shown in the figure below. The coefficient of friction between the cylinder and the plank is 0.5 . The plank is given a velocity of 20 m/s toward right. Find the time (in seconds) after which the plank and cylinder will separate. ($g = 10 \text{ m/s}^2$)



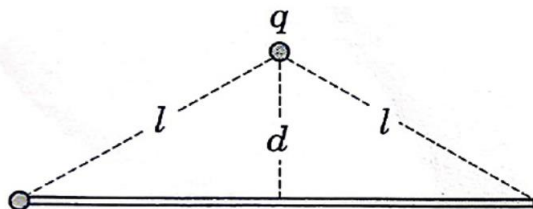
15. A uniform solid spherical ball starts from rest on a loop-the-loop track. It rolls without slipping along the track. However, it does not have enough speed to make it to the top of the loop. From what height h would the ball need to start in order to land at point P directly underneath the top of the loop? Express your answer in terms of R , the radius of the loop. Assume that the radius of the ball is very small compared to the radius of the loop, and that there are no energy losses due to friction.



ELECTROSTATICS

16. A ring with a radius of 0.1 m is made of a thin metallic wire with a cross-sectional area of 10^{-6} m^2 . The ring has a uniform charge of $\pi \text{ C}$. A charge of 10^{-8} C is placed at the center of the ring. Then (Consider that Young's modulus of the metal is $2 \times 10^{11} \text{ N m}^{-2}$.)
- A the tension in the wire is $1.5 \times 10^2 \text{ N}$.
- B the tension in the wire is $4.5 \times 10^3 \text{ N}$.
- C the change in the radius of the ring is $1.25 \times 10^{-4} \text{ m}$.
- D the change in the radius of the ring is $2.25 \times 10^{-3} \text{ m}$.
17. A fixed circle of radius a and center O is drawn. A charge e is placed at a distance $\frac{3a}{4}$ from O on a line through O perpendicular to the plane of the circle. Then, a second charge e' is similarly placed at a distance $\frac{5a}{12}$ on the opposite side of the circle and there is no net flux through the circle. Then,
- A the flux through the circle due to charge e is $\frac{e}{5\epsilon_0}$.
- B the value of e' is $\frac{13e}{20}$.
- C the value of e' is $\frac{80e}{91}$.
- D e' is greater than e .
18. A point like positive charge q is fixed at a distance d on the perpendicular bisector of thin electrically neutral nonconducting rod. Ends of the rod are at distance l from the point charge. A bead of mass m having the same charge q is threaded on the rod. The bead is projected from one end of the rod towards the other end. Velocity of the bead at the centre of the rod is observed to be v_1 and at the other end v_2 . coefficient of friction between the bead and the rod is uniform everywhere, if velocity of projection of the bead is

$$\sqrt{av_1^2 + bv_2^2 + \frac{cq^2}{4\pi\epsilon_0 m} \left(\frac{1}{d} - \frac{1}{l} \right)} \quad \text{find } a+b+c ?$$



19. A particle of charge Q is placed on the axis of a neutral cylinder of a volume V at a distance x from one end of the cylinder as shown in figure. Here the distance x is much larger than the linear dimensions of the cylinder. F is the electrostatic interaction between the charge particle and the cylinder and q is the magnitude of charge induced on the cylinder flat face of area A , then (Use suitable approximations)

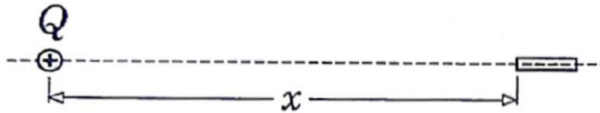
(a) If cylinder is of conducting material $F = \frac{Q^2 V}{8\pi^2 \epsilon_0 x^5}$

(b) If cylinder is of conducting material $q = \frac{QA}{4\pi x^2}$

(c) If cylinder is of insulating material of relative permittivity ϵ_r

$$F = \frac{Q^2 V (\epsilon_r - 1)}{8\pi^2 \epsilon_0 x^5 \epsilon_r}$$

(d) If cylinder is of conducting material $F = \frac{Q^2 V}{16\pi^2 \epsilon_0 x^5}$

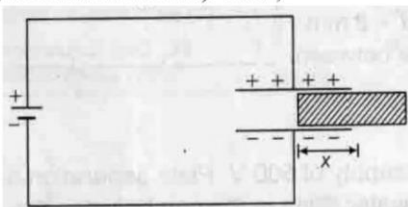


20. A non-conducting ring of mass m and radius r is lying at rest in the vertical xy plane on a smooth non-conducting horizontal xz plane. Charges $+q$ and $-q$ are distributed uniformly on the ring on the two sides of the vertical diameter of the ring. A constant and uniform electric field $\vec{E}$ is set up along the x -direction. The ring is given a small rotation about the vertical diameter of the ring and released. the time period of oscillation of the

ring is $a\pi \sqrt{\frac{\pi m r}{b q E}}$, where b^a is

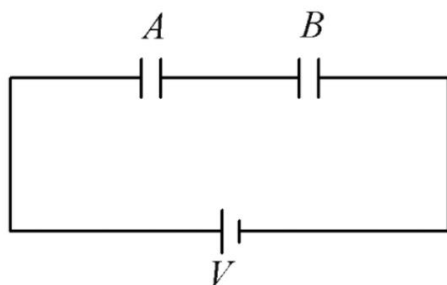
CAPACITORS

21. A capacitor is charged by using a battery and then a dielectric slab is released from the position shown in the figure (battery remains connected). Then,



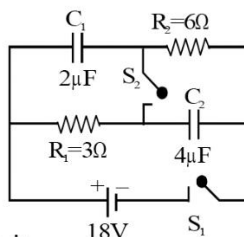
- (a) force on dielectric slab does not depend on the dimension x
 (b) force on dielectric slab depends on the dimension x
 (c) Time period of oscillation does not depend on the dimension x
 (d) time period of oscillation depends on the dimension x

22. Two identical parallel plate capacitors A and B are connected in series through a battery of emf V volt, as shown in the Figure. Area of each plate is a and initial distance of separation of plates in each capacitor is d . Now, the separation in B starts increasing at constant rate v . When the separation of the plates in B is $2d$, then the rate of work done on the battery is $\frac{a\epsilon_0 v V^2}{nd^2}$. What is the value of n ?



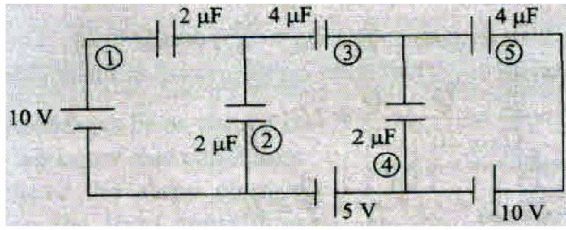
23. A parallel plate capacitor is lowered into water ($\epsilon_r = 81$) in a horizontal position, with water filling up the gap between the plates separated by $d = 1$ mm. Then a constant voltage of 500 V is applied to the capacitor. Find the pressure increment in the gap.

24. In the circuit shown initially the switches are open and capacitors are uncharged. Switches S_1 and S_2 are closed simultaneously at $t = 0$.



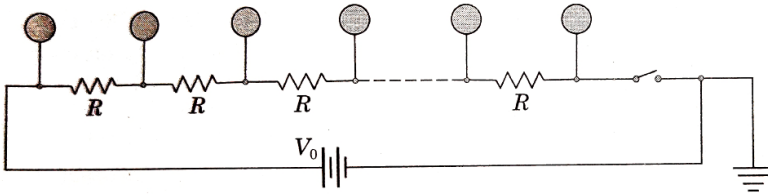
- Charge on capacitor C_2 in steady state is
 - $12 \mu\text{C}$
 - $24 \mu\text{C}$
 - $48 \mu\text{C}$
 - none of these
- Now switch S_2 is opened after a long time. Then, charge flow through the S_1 is
 - $12 \mu\text{C}$
 - $24 \mu\text{C}$
 - $36 \mu\text{C}$
 - $48 \mu\text{C}$
- In above question, the amount of heat dissipated in resistors is
 - $136 \mu\text{J}$
 - $272 \mu\text{J}$
 - $68 \mu\text{J}$
 - $216 \mu\text{J}$

25. In the figure shown, what are the charges on all the five capacitors



CURRENT ELECTRICITY

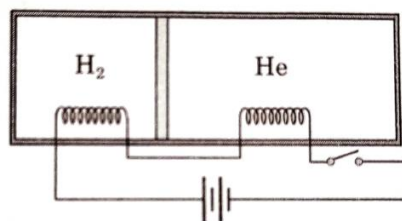
26. Resistors $R, 2R, 3R, \dots, 100R$ are connected in series and free ends of this series are connected to make a closed loop. An ideal battery is connected between the junction of resistor R and $100R$ and the junction of resistors nR and $(n+1)R$. Find value of n for which a minimum current flows through the battery?
27. The given circuit consists of an ideal battery, n identical resistors each of resistances R and $n+1$ identical conducting spheres each of radius r . Assume the balls to be at great distances from each other as well from the circuit. The total charge finally accumulated on all the spheres after the key is closed is $a\pi\epsilon_0 r(n+b)V_0$, find $a+b$



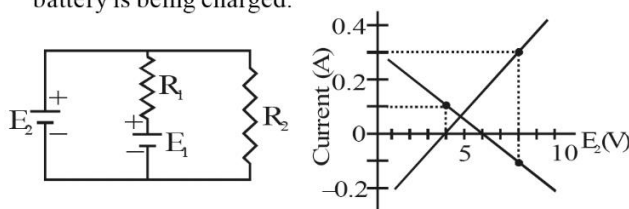
28. H is amount of heat generated in a resistor of resistance R due to charge q passing through it. If the current in the resistor
- decreases down to zero uniformly during a time interval t_0
then, $H = \frac{4 q^2 R}{3 t_0}$
 - decreases down to zero uniformly during a time interval t_0
then, $H = \frac{2 q^2 R}{3 t_0}$
 - decreases down to zero with its value becoming half after every t_0 seconds then, $H = \frac{q^2 R \ln 2}{2 t_0}$
 - decreases down to zero with its value becoming half after every t_0 seconds then, $H = \frac{q^2 R \ln 2}{3 t_0}$

29. A container thermally insulated from the surroundings is divided into two parts by a heat-insulating piston that can slide in the container without friction. On one side of the piston, mass m_1 of gaseous hydrogen occupies volume V_1 and on the other side mass m_2 of gaseous helium occupies volume V_2 . Molar masses of these gases are M_{H_2} and M_{He} respectively. The gases can be heated by two series connected heating coils. The resistance of the coil which heats hydrogen gas depends on absolute temperature T according to the law $R_1 = R_0 + AT$. the piston does not move after closing the switch. If the law describing dependence of resistance of the coil heating the helium gas is $\left(R_0 + AT \frac{m_2 M_{H_2} V_1}{m_1 M_{He} V_2} \right) \left(\frac{aV_2}{bV_1} \right)$, find $a+b$

The gases always remain in thermal equilibrium with the heating coils.



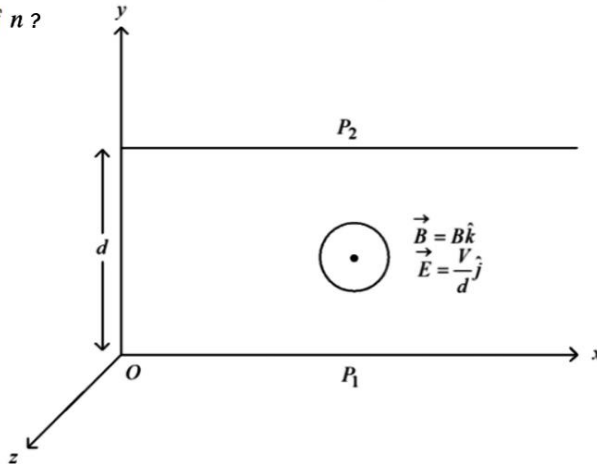
30. In the circuit shown, both batteries are ideal. Emf E_1 of battery 1 has a fixed value, but emf E_2 of battery 2 can be varied between 1 V and 10V. The graph gives the currents through the two batteries as a function of E_2 , but are not marked as to which plot corresponds to which battery. But for both plots, current is assumed to be negative when the battery is being charged.



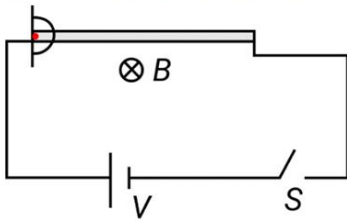
1. The value of emf E_1 is
 - (a) 8V
 - (b) 6V
 - (c) 4V
 - (d) 2V
2. The resistance R_1 has value
 - (a) 10 Ω
 - (b) 20 Ω
 - (c) 30 Ω
 - (d) 40 Ω
3. The resistance R_2 is equal to
 - (a) 10 Ω
 - (b) 20 Ω
 - (c) 30 Ω
 - (d) 40 Ω

MOVING CHARGES & MEC

31. A potential difference of V volt is applied between two parallel plates P_1 and P_2 , separated by a distance d , with the planes of plates being parallel to $x-z$ plane as shown in the Figure. A uniform magnetic field $\vec{B} = B\hat{k}$ is applied out of the cross-section, perpendicular to it. A particle of charge q and mass m is released at rest from $O(0, 0, 0)$ on the plate P_1 . It is found that the particle never touches the plate P_2 if $d = \frac{1}{B} \sqrt{\frac{nVm}{q}}$. What is the minimum value of n ?

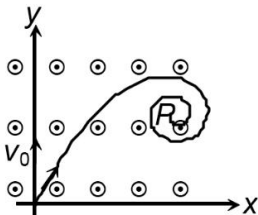


32. A uniform conducting rod of length l and mass m is hinged about one of its ends and is resting on a conducting support, as shown in the figure below. Initially, the rod is horizontal. A battery of emf V is connected across the rod as shown. A magnetic field B is present perpendicular to the plane of paper. The switch is closed at $t=0$. The rod is just able to reach the vertical position. The whole circuit is resistance-less. Then



- A the charge that flows through the rod is $\frac{2m}{3B} \sqrt{\frac{3g}{l}}$.
- B the angular velocity of the rod at the moment of leaving the contact is $\sqrt{\frac{3g}{l}}$.
- C the rod experiences a net force in the vertical upward direction, just after closing the switch.
- D the rod will remain at rest just after closing the switch.

33. A particle having mass m , charge q is projected with velocity v_0 along y -axis in a region of uniform magnetic field B which is outward and perpendicular to the plane of the paper as shown in the figure. The particle is continuously subjected to a frictional force which varies with velocity as $\vec{F}_r = -\alpha \vec{v}$, where α is a constant. Consequently the particle moves on a spiral path till it comes to rest at point P .



1. The speed of the particle as a function of time is

- (a) $v = v_0 e^{-\alpha t/m} \cos \frac{qB}{m} t$ (b) $v = v_0 e^{-\alpha t/m}$
 (c) $v = v_0 e^{-2\alpha t/m} \cos \frac{qB}{m} t$
 (d) $v = v_0 e^{-2\alpha t/m} \left(1 - \cos \frac{qB}{m} t \right)$

2. The x component of velocity as a function of time is

- (a) $v_0 e^{-\alpha t/m} \sin \frac{qB}{m} t$ (b) $v_0 e^{-2\alpha t/m} \sin \frac{qB}{m} t$
 (c) $v = v_0 e^{-\alpha t/m} \left(1 - \cos \frac{qB}{m} t \right)$
 (d) $v_0 e^{-2\alpha t/m} \left(1 - \cos \frac{qB}{m} t \right)$

3. The x coordinate of point P is

- (a) $\frac{2qBmv_0}{\alpha^2 + q^2 B^2}$ (b) $\frac{qBmv_0}{\alpha^2}$
 (c) $\frac{mv_0}{qB}$ (d) $\frac{qBmv_0}{\alpha^2 + q^2 B^2}$

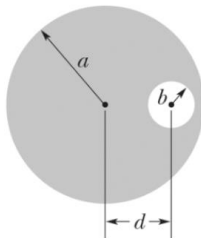
34. The space has electromagnetic field which varies with time whose variation is given as

$$\vec{B} = \begin{cases} -B_0 \hat{k} & \text{if } 0 \leq t \leq \frac{\pi m}{qB_0} \\ -B_0 \hat{j} & \text{if } \frac{\pi m}{qB_0} \leq t \leq \frac{2\pi m}{qB_0} \\ 0 & \text{if } \frac{2\pi m}{qB_0} \leq t < \infty \end{cases} \text{ and } \vec{E} = \begin{cases} 0 & \text{if } 0 \leq t \leq \frac{2\pi m}{qB_0} \\ -E_0 \hat{k} & \text{if } \frac{2\pi m}{qB_0} \leq t < \infty \end{cases}$$

A charge particle having mass m and positive charge q is given velocity $v_0 \hat{i}$ at origin at $t = 0$ sec. The coordinate of point on xy plane when it again passes through xy plane for the first time is :

- (A) $\left(\frac{2mv_0}{q} \sqrt{\frac{v_0}{E_0 B_0}}, \frac{2mv_0}{qB_0}, 0 \right)$ (B) $\left(\frac{2mv_0}{q} \sqrt{\frac{v_0}{E_0 B_0}}, \frac{mv_0}{qB_0}, 0 \right)$
 (C) $\left(\frac{mv_0}{q} \sqrt{\frac{v_0}{E_0 B_0}}, \frac{2mv_0}{qB_0}, 0 \right)$ (D) $\left(\frac{mv_0}{q} \sqrt{\frac{v_0}{E_0 B_0}}, \frac{mv_0}{qB_0}, 0 \right)$

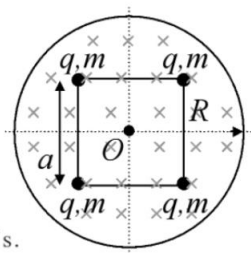
35. Figure shows a cross section of a long cylindrical conductor of radius $a = 4.00$ cm containing a long cylindrical hole of radius $b = 1.50$ cm. The central axes of the cylinder and hole are parallel and are distance $d = 2.00$ cm apart; current $i = 5.25$ A is uniformly distributed over the tinted area. Then,



- (a) B at the center of the hole is 1.53×10^{-5} T
 (b) B at the center of the hole if $b = 0$ is $B = \frac{\mu_0 i d}{2\pi a^2}$
 (c) B at the center of the hole if $d = 0$ is non-zero
 (d) B at the center of the hole if $d = 0$ is zero

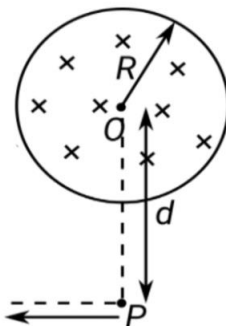
EMI & AC

36. Four identical charge particles each of mass 0.1 kg and charge 2 C connected to each other via massless non-conducting rods of equal length. The whole arrangement is placed in a cylindrical region carrying a uniform magnetic field as shown in the figure ($B_0 = 4$ T, $a = 1$ m). Suddenly the magnetic field is switched off. Then choose the correct statement(s).



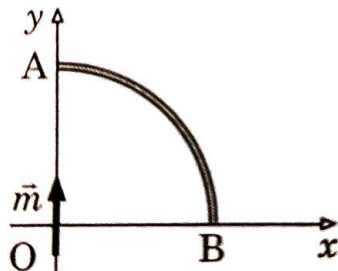
- A The angular momentum of the system is $8 \text{ N} \cdot \text{m/s}$.
 B The angular momentum of the system is $8\sqrt{2} \text{ N} \cdot \text{m/s}$.
 C The angular velocity of the system is 40 rad/s .
 D The magnetic field at point O after the magnetic field is switched off is non zero.

37. In a cylindrical region of radius R , there exists a time varying magnetic field B such that $\frac{dB}{dt} = k$ (> 0). A charged particle having charge q is placed at the point P at a distance d ($> R$) from its centre O . Now, the particle is moved in the direction perpendicular to OP (see figure) by an external agent upto infinity so that there is no gain in kinetic energy of the charged particle. Choose the correct statement/s.

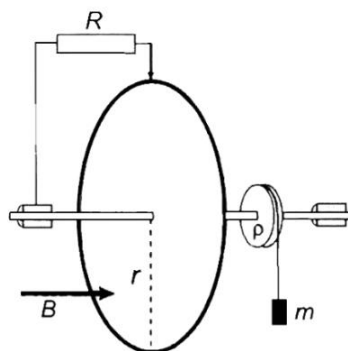


- A Work done by external agent is $\frac{q\pi R^2}{4} k$ if $d = 2R$.
 B Work done by external agent is $\frac{q\pi R^2}{8} k$ if $d = 4R$.
 C Work done by external agent is $\frac{q\pi R^2}{4} k$ if $d = 4R$.
 D Work done by external agent is $\frac{q\pi R^2}{4} k$ if $d = 6R$

38. Consider a quarter circular conducting ring of large radius r with its centre at the origin, where a magnetic dipole of moment $\vec{m}$ is placed as shown in the figure. If the ring rotates at a constant angular velocity ω about the y -axis, electromotive force induced between its ends is
- (a) Zero
(b) $\mu_0 m \omega / (2\pi r)$
(c) $\mu_0 m \omega / (4\pi r)$
(d) $\mu_0 m \omega / (8\pi r)$



39. A disc of radius r is made of a material of negligible resistance and can rotate about a horizontal shaft. A smaller disc of radius ρ is fixed onto the same shaft and has a massless cord wrapped around it, which is attached to a small object of mass m as shown. Two ends of a resistor of resistance R are connected to the perimeter of the disc and to the shaft by wiping contacts. The system is then placed into a uniform horizontal magnetic field B and mass m is released. Find the constant angular velocity with which the disc will rotate after a certain time. Data: $r = 10 \text{ cm}$, $\rho = 2 \text{ cm}$, $R = 0.01 \Omega$, $B = 0.2 \text{ T}$, $m = 50 \text{ g}$.



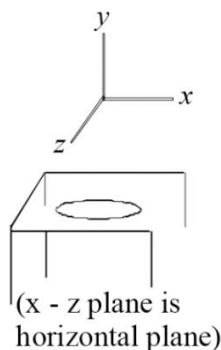
40. A uniform conducting ring of mass $\pi \text{ kg}$ and radius 1 m is kept on smooth horizontal table. A uniform but time varying magnetic field $B = (\hat{i} + t^2 \hat{j})$ Tesla is present in the region. (t is time in seconds, resistance of ring is 2Ω , $g = 10 \text{ m/s}^2$). Then,

- Time (in second) at which ring start toppling is

(a) $10/\pi$ (b) $20/\pi$
(c) $5/\pi$ (d) $25/\pi$
- Heat generated (in kJ) through the ring till the instant when ring start toppling is

(a) $1/3\pi$ (b) $2/\pi$
(c) $2/3\pi$ (d) $1/\pi$
- Induced electric field (in volt/meter) at the circumference of ring at the instant ring start toppling is

(a) $10/\pi$ (b) $20/\pi$
(c) $5/\pi$ (d) $25/\pi$

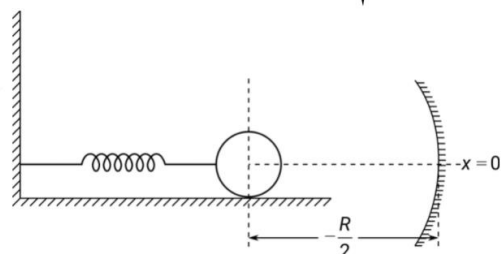


RAY OPTICS

41. In the arrangement shown in the figure below, a small spherical object with a mass of m is at rest on a smooth floor at $x = -\frac{R}{2}$ and is connected to a spring with a force constant of k . At $t = 0$ s, it is given a velocity

$$\vec{V} = \frac{R}{4} \sqrt{\frac{k}{m}} \hat{i}$$

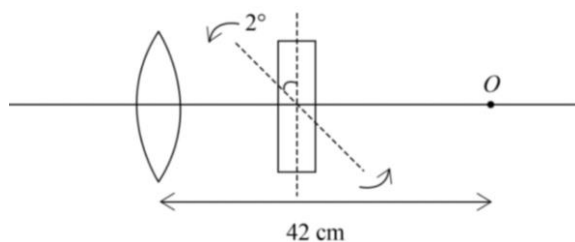
The speed of the image of this object in the stationary concave mirror at time $t' = \frac{3\pi}{4} \sqrt{\frac{m}{k}}$ is



- (A) $\sqrt{2}R\sqrt{\frac{k}{m}}$
 (B) $R\sqrt{\frac{k}{m}}$
 (C) $\frac{R}{4}\sqrt{\frac{k}{m}}$
 (D) zero

(R is radius of curvature of concave mirror)

42. An object is placed at a distance of 42 cm from a convex lens of focal length 20 cm, if a glass slab of thickness 4 cm and refractive index 2 is introduced in between the object and the lens then image is formed at I_1 , if now the slab is tilted by an angle of 2° in the sense as shown in the figure then the new position of image is I_2 , the distance between I_1 & I_2 is $\frac{x}{90}$ cm. Find ' x '. (consider paraxial approximation)



43. The flat surface of a plano-convex lens is silvered and cut to give the shape of a cross-section as ABCD. The optical centre of the silvered lens is attached to a rigid support through a spring of force constant k . The radius of curvature of BC is 10 cm and the refractive index of the material of the lens is 1.5. The lens is initially at rest so that the elongation of the spring is x_0 . A particle of mass $m = \frac{kx_0}{g}$ hits the lens elastically with velocity 10 m/s as shown in the figure. Taking $x_0 = \frac{0.1}{\pi^2}$ m and $g = 10 \text{ m/s}^2$.

1. The equivalent focal length of the silvered lens is

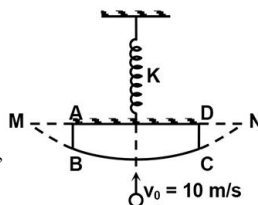
- (A) -5 cm (B) -10 cm
 (C) -15 cm (D) -20 cm

2. Let $t = 0$ be the time of collision, then at time $t = \pi \sqrt{\frac{m}{k}}$, the speed of the image of the particle relative to the lens is

- (A) 40 m/s (B) 36 m/s
 (C) 32 m/s (D) none of these.

3. If the lens were complete as MBCNM, then

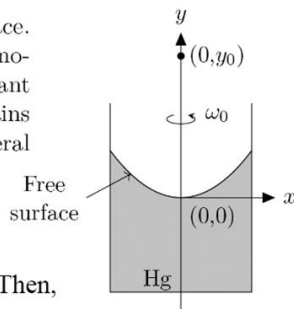
- (A) the image velocity will be more (B) the image velocity will be less
 (C) the image will be more intense (D) can't be predicted



44. The free surface of mercury (Hg) is a good reflecting surface. A tall cylinder partly filled with Hg and possessing total moment of inertia I is rotated about its axis with the constant angular velocity ω_0 as shown in figure. The Hg surface attains a paraboloidal profile. The radius of curvature ρ of a general profile is given by

$$\rho = \left| \frac{[1 + (dy/dx)^2]^{3/2}}{d^2y/dx^2} \right|$$

Consider a point object at $(0, y_0)$ as shown in the figure. Then,



- (A) $\rho_{x=0} \approx 15$ cm when $\omega_0 = 78$ rpm
 (B) For real images of point object $y_0 > g/2\omega_0^2$ (consider paraxial rays)
 (C) For real images of point object $y_0 < g/2\omega_0^2$ (consider paraxial rays)
 (D) For virtual images of point object $y_0 < g/2\omega_0^2$ (consider paraxial rays)

45. There is a hole in the middle of small thin circular converging lens of focal length $f = 4$ cm. The diameter of the hole is half of the diameter of the lens. There is a pointlike light-source $A = 9$ cm away from a wall.

Lens can be placed at a distance of x or y from light source in order to get single, circular illuminated bright spot on the wall, which also has a sharp edge. Then the value of $x+y$ (in cm) is ,
 (x or y is not equal to zero)

MODERN PHYSICS

46. A particle of mass m moves along a circular orbit in a centro - symmetrical field of potential $U(r) = \frac{\alpha r^2}{2}$ where α is a constant and r is the radius of the circle. If the particle obeys Bohr's quantization condition, then

the allowed energy levels are $E_n = n\hbar\omega$ where

A $\omega = \sqrt{\frac{\alpha}{m}}, n = 1, 2, 3, \dots$

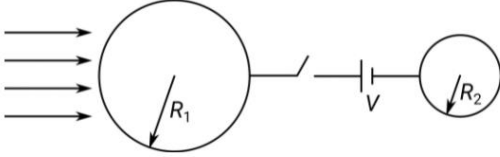
B the radii of allowed orbits is given by $r_n = \sqrt{\frac{n\hbar}{m\omega}}, n = 1, 2, \dots$

the wavelength of the photon that is emitted from this system

C with the least energy is given by $\lambda = 2\pi c \sqrt{\frac{m}{\alpha}}$.

D the central force holding the particle on the circular orbit is proportional to the radius r .

47. Two conducting uncharged spheres of radius R_1 and R_2 ($R_1 > R_2$) are connected to a battery with a switch as shown. Light rays of frequency f are incident on the bigger sphere. Simultaneously, the switch is closed. The work function of the bigger sphere is ϕ . After some time, the charge on bigger sphere is q_1 , and on the smaller sphere is $-q_2$. It remains constant thereafter. Then

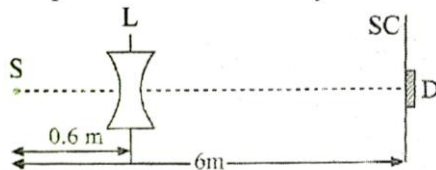


- A $V = \frac{q_1}{4\pi\epsilon_0 R_1} + \frac{q_2}{4\pi\epsilon_0 R_2}$
- B The number of electrons emitted by the larger sphere is $\frac{q_1 - q_2}{e}$ when e is the charge of an electron.
- C $\frac{eq_1}{4\pi\epsilon_0 R_1} = hf - \phi$
- D $\frac{eq_1}{\pi\epsilon_0 R_1} = hf - \phi$

48. A radioactive nuclide X is produced at a constant rate α . It decays into another element with decay constant λ and during each decay, E_0 energy is released. If 50% of this energy is utilised in melting ice at 0°C , the mass of ice that will melt in one mean life is $\frac{\alpha E_0}{n\lambda eL}$, where n is

(At $t = 0$, number of nuclei of X is zero and latent heat of fusion of ice is L)

49. A monochromatic point source S of power 2W is radiating light of wavelength $6000\text{\AA}$. A concave lens L of focal length 0.6m and aperture 0.1m is placed at a distance of 0.6m from the source as shown. A photoemissive detector D of surface area 0.5cm^2 is placed at the centre of the screen SC at a distance of 6m from the source. The efficiency of the detector of the photoelectrons generated per incident photon is 0.9 . Assume a uniform illumination on the screen and a transmission of 80% from the lens. If photon flux and photocurrent detected by detector is x and y , then



- a) $x = 1.33 \times 10^{11} \text{ s}^{-1}$
- b) $x = 3.45 \times 10^{11} \text{ s}^{-1}$
- c) $y = 2.13 \times 10^{-8} \text{ A}$
- d) $y = 4.5 \times 10^{-8} \text{ A}$

50. A neutron collides head on with a stationary H-atom in its ground state. Assume that the mass of neutron is same as that of H-atom and kinetic energy before collision is K . Then,
- (a) If $K = 12$ eV implies collision is elastic
 - (b) If $K = 20.4$ eV implies collision is either elastic or partially elastic
 - (c) If $K = 21$ eV implies collision is either elastic or partially elastic
 - (d) If $K = 24.18$ eV implies collision is elastic or partially elastic or perfectly inelastic



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